

Makda T. Mammo

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PROFILE

I am a 5th-year Software Engineering student at Mekelle University with a technical foundation in backend systems and machine learning. My focus is now on the governance, ethics, and societal impact of such technologies. I aim to apply my technical insight to analyze, evaluate, and help formulate policies for equitable and accountable digital ecosystems, with a specific concern for vulnerable and post-conflict communities.

EDUCATION

B.Sc. in software engineering CGPA 3.96/4.0

Mekelle University 

09/2019 – 07/2026

Mekelle, Ethiopia

Relevant coursework

- Data structures and algorithms, Artificial intelligence, machine learning, cloud computing, compiler design, project management, web design and programming

Coding Bootcamp

addiscoder 

07/2019 – 08/2019

Addis Ababa, Ethiopia

- Python, data structures, and algorithms
- 4-week intensive program that covered everything from simple print statements to complex topics like dynamic programming, memoization, and graph algorithms

PROFESSIONAL EXPERIENCE

Buna Labs

Backend Engineer

01/2025 – Present

Addis Ababa, Ethiopia

- Designed and implemented backend systems enabling seamless integration with frontend applications
- Built secure authentication systems
- Collaborated with frontend developers to integrate Django backend with React/Vue.js
- Integrated Redis to cache highly accessed profile data, improving scalability and reducing API response time by over 20%.
- I developed a rigorous testing pipeline, authoring over **200 unit and integration tests** using **Pytest**, achieving **85%** test coverage
- **Technologies used:** Python, Django REST Framework, Celery, Redis, JWT, Render, gunicorn,

AddisCoder

Teaching Assistant

07/2025 – 08/2025

Addis Ababa, Ethiopia

- Enhanced student understanding of Data Structures and Algorithms by leading lab sessions for 23 students.
- Improved learning outcomes by creating and refining 4 Jupyter Notebooks daily, offering tailored exercises.
- Professors: Jelani Nelson (UC Berkeley/Google), Daniel Kang (UIUC), Huy Nguyen (Northeastern), Huacheng Yu (Princeton)

Research & Technology Intern

Mozilla Festival - GirlsWhoML Sprintship

11/2025 – 12/2025

Remote

- Selected for a competitive research sprintship analyzing the societal impact of machine learning, with a focus on bias, fairness, and disparate effects on women and marginalized communities.
- Contributed to the Mozilla Festival 2025 under its theme "**Unlearn the Default,**" participating in the GirlsWhoML research sprintship.
- Conducted collaborative research and primary data collection on topics including algorithmic fairness, dataset representation, and real-world model evaluations.
- Participated in structured virtual workshops (via Zoom) focused on developing frameworks and resources to translate technical ML challenges into actionable insights for more equitable and responsible technology policy.

SKILLS

Languages

- Python, java, C++, C#

Frameworks

- Django, Django REST Framework

Backend Tools

- supabase, firebase, celery, redis, postman

Cloud & DevOps

- AWS, Docker, Heroku, GitHub Actions, Gunicorn, render

Soft Skills

- Teamwork, communication, leadership

PROJECTS

MentorMe

10/2025 – Present

- Designed and developed **MentorMe**, a mobile platform that connects tutors, students, and parents to address information gaps in informal education markets.
- Built a **Flutter** client with role-based authentication and separate workflows for tutors, learners, and parents.
- Implemented a **hybrid recommendation system** combining content-based filtering with matrix factorization to match tutors and learners based on profiles and interaction patterns.
- Integrated **real-time chat and tutoring request tracking** to model and manage tutoring relationships.
- **Technologies:** Flutter (Dart), Firebase Authentication, Cloud Firestore, Matrix Factorization, Content-Based Filtering, Real-Time Messaging

Talent Search

07/2025

- Collaborated with frontend and backend developers to design and implement Talent Search, a platform connecting young talents with verified employers.
- Developed secure RESTful APIs using Django REST Framework and PostgreSQL for efficient data storage.
- Optimized performance with Redis caching, reducing API response times by over 20%.
- Wrote unit and integration tests with Pytest, improving production reliability.
- **Technologies:** Python, Django REST Framework, PostgreSQL, Redis, JWT, Pytest, Gunicorn, Render, Git/GitHub

Tirumba

08/2025 – Present

- Designed and developed Tirumba, a digital funeral announcement system that could replace manual loudspeaker broadcasts in Tigray.
- Built a Django backend with Celery & Redis to handle scalable SMS and push notification dispatch.
- Integrated Twilio SMS into Tirumba to manage user registration and automated funeral notifications
- **Technologies:** Django REST Framework, PostgreSQL, Celery, Redis, Twilio API, Firebase, Docker, Nginx

CERTIFICATES

GirlsWhoML Research and Technology Internship Completion

12/2025

AddisCoder TA

08/2025

100 Days of Code: The Complete Python Pro Bootcamp

07/2025

- Content: Python basics, OOP, Web development, Data Science and Machine Learning
- Instructor: Dr. Angela Yu

AddisCoder Student

08/2019